

Ahsanullah University of Science and Technology

Department of Computer Science and Engineering

Term Assignment

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A report on

Cardiovascular Disease Prediction Using Machine Learning Techniques

Course Number: CSE 3208

Course Title: Introduction to Artificial Intelligence Lab

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Problem Statement

This project aims to predict the presence of cardiovascular disease in patients using machine learning classification models. Using a dataset with various health metrics and lifestyle factors, the goal is to classify individuals as either having cardiovascular disease (cardio = 1) or not having it (cardio = 0). Two classification models, Decision Tree and K-Nearest Neighbors (KNN), were implemented and compared to determine which performs better for this prediction task.

Dataset Overview

The dataset used in this analysis, `health_data.csv`, contains 70,000 records with the following attributes:

- **age**: Patient's age
- **gender**: Binary (0 = female, 1 = male)
- **height**: Height in centimeters
- **weight**: Weight in kilograms
- **ap_hi**: Systolic blood pressure
- **ap_lo**: Diastolic blood pressure
- **cholesterol**: Cholesterol level (0 = normal, 1 = above normal, 2 = high)
- **gluc**: Glucose level (0 = normal, 1 = above normal, 2 = high)
- **smoke**: Smoking status (0 = no, 1 = yes)
- **alco**: Alcohol consumption (0 = no, 1 = yes)
- **active**: Physical activity level (0 = no, 1 = yes)
- **cardio**: Target variable - presence of cardiovascular disease (0 = no, 1 = yes)

Data Preprocessing

The following preprocessing steps were applied to prepare the data for modeling:

1. The dataset was loaded and examined for missing values, with no missing values found.
2. A correlation matrix was created to identify potential relationships between features.
3. The data was split into training (67%) and testing (33%) sets using a fixed random seed of 42 for reproducibility.
4. Features were standardized using `StandardScaler` to ensure all variables contributed equally to the model, particularly important for the KNN algorithm.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33,
random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Model Implementation

Decision Tree Classifier

A Decision Tree model was implemented with default parameters and a fixed random state for reproducibility:

```
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict(X_test)
```

K-Nearest Neighbors (KNN) Classifier

A KNN model was implemented with 5 neighbors and trained on the standardized data:

```
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train_scaled, y_train)
y_pred_knn = knn_model.predict(X_test_scaled)
```

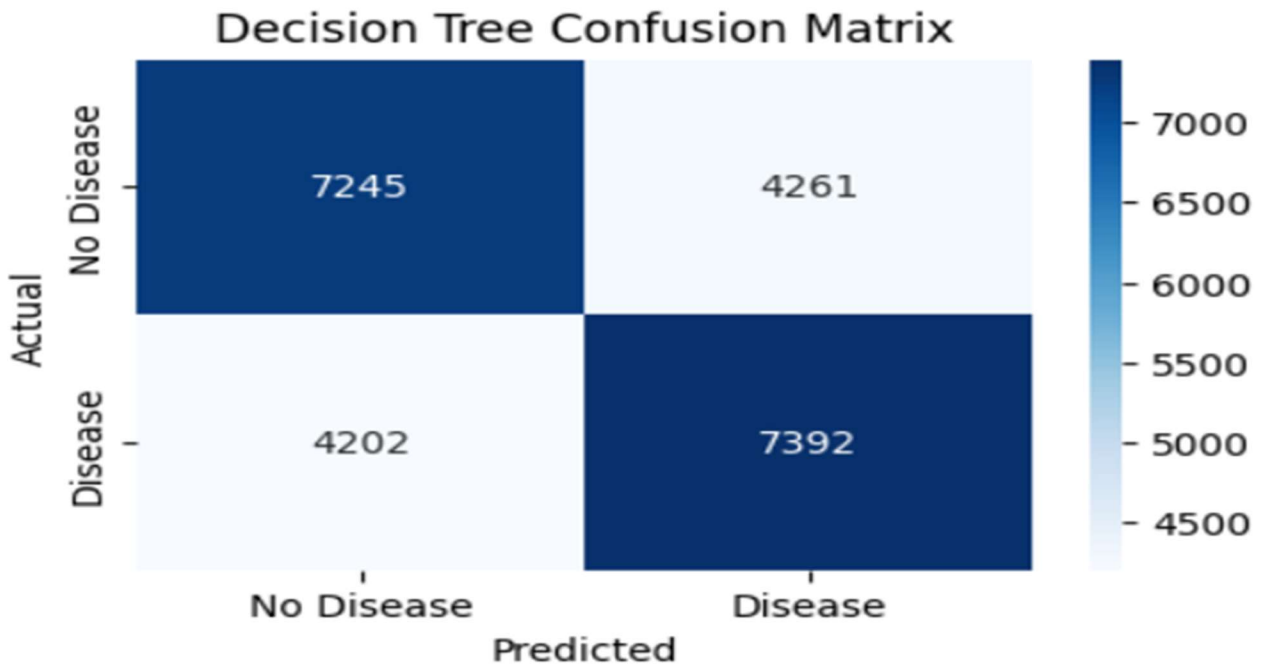
Model Evaluation

Both models were evaluated using multiple performance metrics:

1. **Accuracy:** Proportion of correctly classified instances
2. **Precision:** Ratio of correctly predicted positive observations to total predicted positives
3. **Recall:** Ratio of correctly predicted positive observations to all actual positives
4. **F1-score:** Harmonic mean of precision and recall

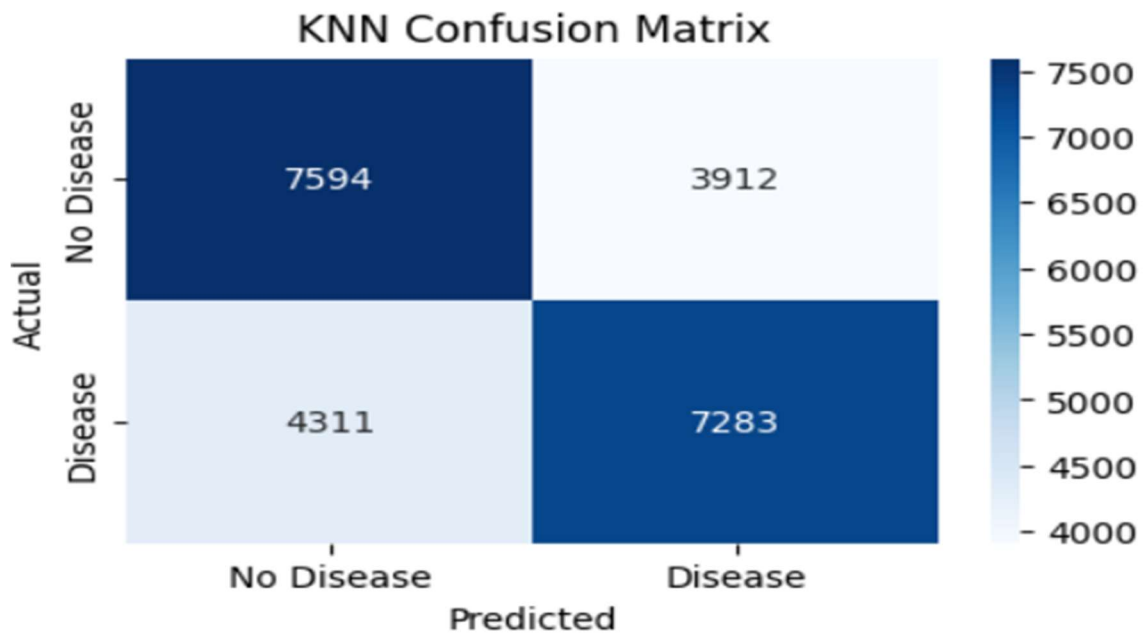
Decision Tree Results

- Accuracy: 0.7273
- Precision: 0.7136
- Recall: 0.7545
- F1-score: 0.7335



KNN Results

- Accuracy: 0.7321
- Precision: 0.7307
- Recall: 0.7344
- F1-score: 0.7325



Model Comparison

When comparing both models based on their F1-scores:

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.7273	0.7136	0.7545	0.7335
KNN	0.7321	0.7307	0.7344	0.7325

Based on the F1-score, the Decision Tree model (0.7335) marginally outperforms the KNN model (0.7325), although the difference is very small. This suggests that both models have comparable performance for this specific task.

The detailed classification reports for both models provide additional insights into the performance for each class:

Decision Tree Classification Report

	precision	recall	f1-score	support
No Disease	0.74	0.70	0.72	11839
Disease	0.71	0.75	0.73	11560
accuracy			0.73	23399
macro avg	0.73	0.73	0.73	23399
weighted avg	0.73	0.73	0.73	23399

KNN Classification Report

	precision	recall	f1-score	support
No Disease	0.73	0.73	0.73	11839
Disease	0.73	0.73	0.73	11560
accuracy			0.73	23399
macro avg	0.73	0.73	0.73	23399
weighted avg	0.73	0.73	0.73	23399

Conclusion

Both the Decision Tree and KNN models show promising results for predicting cardiovascular disease based on the given health attributes. The Decision Tree model slightly outperforms the KNN model in terms of F1-score, but the difference is minimal.

The Decision Tree model has a higher recall, meaning it's better at identifying actual positive cases (patients with cardiovascular disease), which could be particularly valuable in a clinical setting where missing a diagnosis could have serious consequences.

The KNN model has a more balanced precision and recall, making it a consistent performer overall.

For future work, the following could be considered:

1. Feature engineering to create more informative features
2. Hyperparameter tuning to optimize model performance
3. Testing ensemble methods that combine multiple models
4. Addressing class imbalance if present
5. Exploring more advanced algorithms like Random Forest or Gradient Boosting

In a healthcare context, these models could serve as preliminary screening tools to identify patients who might be at risk of cardiovascular disease, allowing for earlier intervention and potentially better outcomes.